STUDENT PERFORMANCE PREDICTION SYSTEM USING MACHINE LEARNING

Abstract

The Student Performance Prediction System is a machine learning application that predicts academic performance based on educational and behavioural factors. The system analyses attributes such as study hours, attendance, learning resources, motivation, internet access, participation in online courses, assignment completion, discussion participation, learning technology use, extracurricular activities, and stress level.

At first, all available features were used during model training. Several machine learning models were trained and tested, including Logistic Regression, Decision Trees, and Random Forests. To enhance the model's reliability and eliminate unnecessary complexity, feature importance analysis and feature selection were conducted. The Decision Tree classifier was determined to be the most accurate after several trials and comparisons of models and was chosen for deployment.

A Streamlit dashboard was created to deliver live predictions and customised recommendations. The project showcases the application of Machine Learning techniques in the field of educational analytics and student performance evaluation.

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1. Introduction

Educational institutions generate large amounts of student-related data that can be utilised to improve academic outcomes. Student performance is influenced by several factors, including attendance, study habits, motivation, assignment completion, and participation in learning activities.

Machine Learning provides an effective approach for analysing such data and identifying patterns that can help predict academic performance. Predictive systems can help students understand the factors that affect their academic outcomes and encourage data-driven improvements.

The objective of this project is to develop a Machine Learning-based system capable of predicting student performance categories and providing recommendations for improvement through an interactive dashboard.

2. Objectives

The main objectives of this project are:

- Predict student academic performance using Machine Learning techniques.
- Compare the performance of multiple classification algorithms.
- Identify the most influential factors affecting student performance.
- Develop a user-friendly dashboard for real-time predictions.
- Provide personalised recommendations based on prediction results.

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3. Dataset Description

The dataset contains educational and behavioural attributes that influence student academic performance.

Features Used

Feature	Description
StudyHours	Daily study hours
Attendance	Attendance percentage
Resources	Learning resources availability
Extracurricular	Participation in extracurricular activities
Motivation	Student motivation level
Internet	Internet access availability
OnlineCourses	Number of online courses completed
Discussions	Participation in discussions

AssignmentCompletion	Assignment completion percentage
EduTech	Use of educational technology
StressLevel	Student stress level

4. Methodology

Data Preprocessing

Before training the models, the dataset was preprocessed using the following steps:

• Data cleaning

• Label encoding of categorical variables

• Feature scaling using StandardScaler

• Train-test split

• Model training and evaluation

Feature Selection

Initially, the model was built on all available data features. Then, during the experimentation phase of the project, an analysis of feature importance was conducted to assess which variables were providing the greatest contribution(s) to predictive performance.

Some of the features that contributed least to performance (i.e., had low impact) were removed during the experiment to improve model performance through better generalisation and reduced unnecessary complexity. Retraining was performed on each model after feature selection, and performance comparisons were then conducted across the models.

The iterative nature of this methodology enabled the development of a more credible and understandable prediction system for users.

Machine Learning Models

The following classification algorithms were trained and evaluated:

1. Logistic Regression
2. Decision Tree Classifier
3. Random Forest Classifier

Workflow

Data Collection

Data Preprocessing

Feature Selection

Model Training

Model Evaluation

Deployment

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5. Results and Model Comparison

The performance of the machine learning models was evaluated using classification accuracy.

Model	Accuracy
Logistic Regression	28.82%
Decision Tree	88.49%
Random Forest	86.40%

Selected Model

Among all evaluated models, the Decision Tree classifier achieved the highest accuracy and provided better interpretability. Therefore, it was selected as the final model for deployment.

Key Findings

• Assignment Completion was the strongest predictor of success.

• Academic Performance was most affected by class attendance.

• Study Hours contributed positively to correct predictions.

• Online Courses and Learning Engagement improved model results.

• Selection of features helped create more reliable and interpretable models.

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6. Challenges and Improvements

During development, several challenges were encountered.

Initially, all features were used for training, leading to inconsistent predictions despite high accuracy scores. Some test cases produced unrealistic outputs, indicating that additional model refinement was necessary.

To address these issues:

• Feature importance analysis was performed.

• Multiple models were compared.

• Feature selection was applied.

• Predictions were tested using various input combinations.

• The Streamlit application was improved through iterative testing.

Deployment challenges were also encountered while hosting the application on Streamlit Cloud. These issues were resolved by correctly configuring project dependencies.

These improvements contributed to the development of a more robust and user-friendly prediction system.

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7. System Implementation

A web-based dashboard is built using Streamlit which contains an interactive user interface for prediction.

Users of the dashboard can:

• Enter academic and behavioral data.

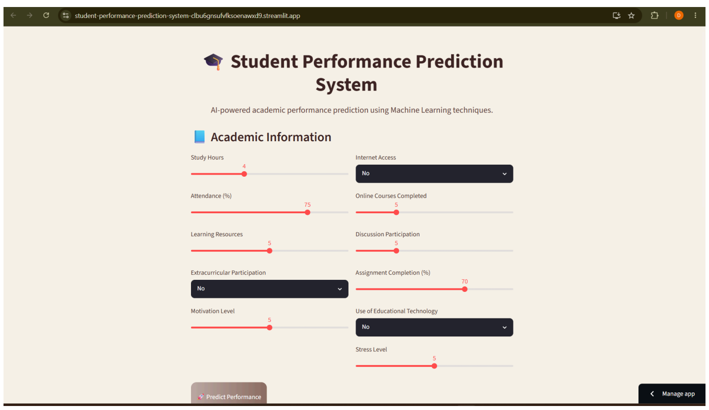
• Instant performance prediction.

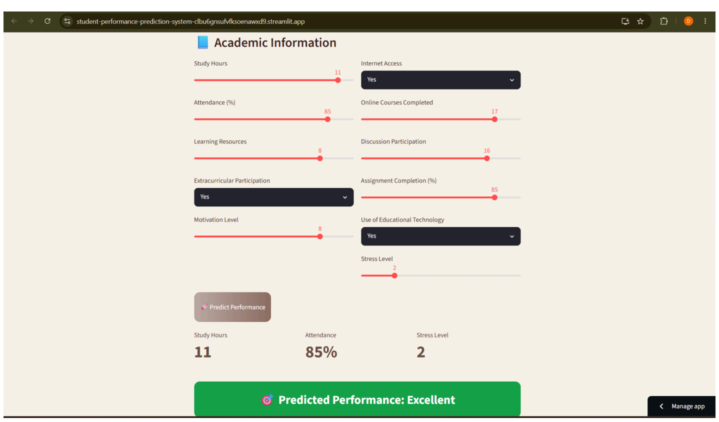
• Get recommendations that are personalized.

• Visualise feature importance.

• Learn about different academic situations.

The system implemented is a live example of Machine Learning in educational analytics.





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8. Conclusion and Future Scope

Conclusion

The project successfully developed Student Performance Prediction System using Machine Learning techniques. Different classification models were trained and compared. The Decision Tree classifier had the best accuracy of 88.49%.

Feature selection and iterative model refinement improved the quality of the predictions and the interpretability of the system. The final solution was deployed using Streamlit and provides real-time performance prediction and personalised recommendations.

The project demonstrates how Machine Learning can be applied effectively in educational analytics to help assess student performance.

Future Scope

• Integration with real educational databases.

• Enhanced recommendation systems.

• Improved visualisation dashboards.

• Institution-wide performance monitoring.

• Cloud-based analytics and reporting.

References

1. Scikit-Learn Documentation
2. Streamlit Documentation
3. Pandas Documentation
4. NumPy Documentation
5. Educational Data Mining Research Articles